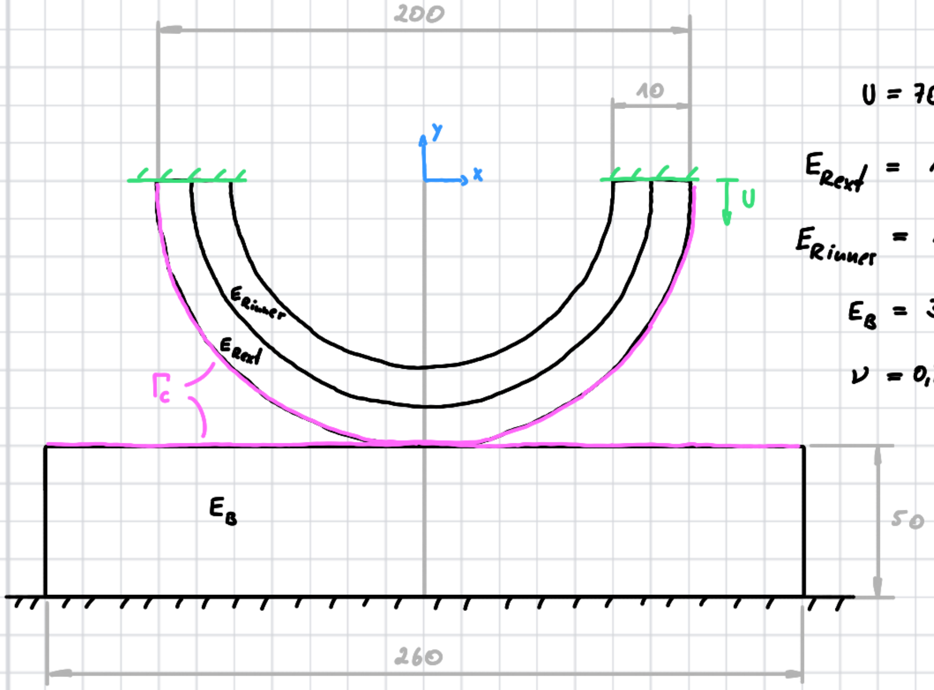
Contact finite strain



$U = 70\,mm$

$E_{Rext} = 1 \cdot 10^3\ N/mm^2$

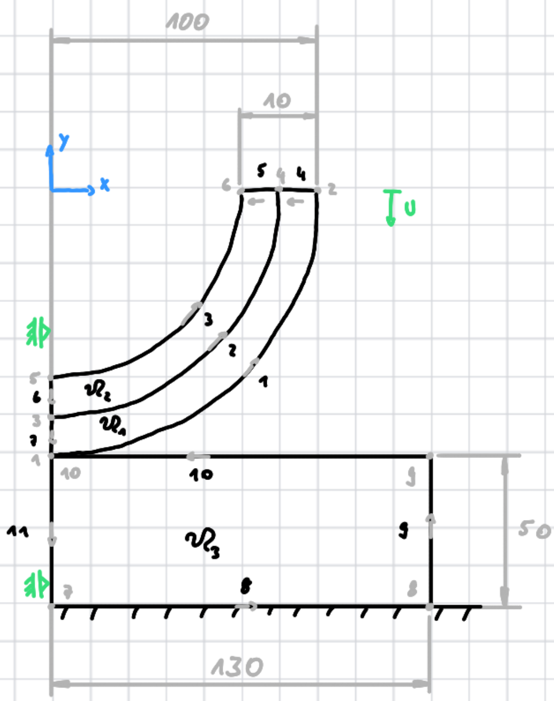
$E_{Rinner} = 1 \cdot 10^5\ N/mm^2$

$E_B = 3 \cdot 10^2\ N/mm^2$

$\nu = 0{,}3$

$U \rightarrow 140$ steps

with $\Delta U = 0{,}5\,mm$



Elements

4, 5, 6, 7 :	1
1, 2, 3 :	32
9, 11 :	10
8, 10 :	26

FEniCSx

$$\Psi = C_{10}(\bar{J}_1 - 3) + \frac{1}{D_1}(J-1)^2$$ Neo-Hook

$$\bar{J}_1 = J^{-2/3} \cdot J_1$$

$$\underline{\underline{F}} = \underline{\underline{I}} + \nabla \underline{u}$$

Plain strain

$$\underline{\underline{F}} = \begin{bmatrix} F_{11} & F_{12} & 0 \\ F_{21} & F_{22} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$J = \det \underline{\underline{F}} \qquad C_{10} = \frac{E}{4(1+\nu)}$$

$$\underline{\underline{C}} = \underline{\underline{F}}^T \underline{\underline{F}} \qquad D_1 = \frac{6(1-2\nu)}{E}$$

$$J_1 = tr\, \underline{\underline{C}}$$

$$\Psi_{tot} = \Psi_1 + \Psi_2 + \Psi_3$$

$$\Pi = \int_{\Omega} \Psi_{tot}\, dx$$

$$D_{\underline{u}} \Pi[\underline{v}] - \int_{\Gamma_c} p_N\, \delta g_N = R \overset{!}{=} 0$$

Discretization

$$\underline{v}_h^T \underline{\underline{K}}\, \underline{u}_h + \underline{v}_h^T \underline{F}_c = \underline{v}_h^T \underline{R}$$

Newton

$$\underline{u}_N^{i+1} = \underline{u}_N^i + \Delta \underline{u}_N^i$$

$$\underline{\underline{J}}_N^i \Delta \underline{u}_N^i = - \underline{R}_N^i$$

Gâteaux derivative $D_x f(x,v) := \frac{d}{dt} f(x + vt)\Big|_{t=0}$

$$J = D_{\underline{u}} R(\underline{u}, \underline{v})[\Delta \underline{u}]$$

$$J = K + K_c$$

$$K_c \approx \int_{\Gamma_c} \varepsilon\, (B_c \Delta \underline{u}_c)(B_c v_c)\, d\Gamma$$

Contact algorithm

$$\underline{x} = \underline{X} + \underline{u}$$

$$N_1 = 1 - \xi$$
$$N_2 = \xi$$

$$\underline{x}^1 = N_1^1(\xi^1) \cdot \underline{x}_1^1 + N_2^1(\xi^1) \cdot \underline{x}_2^1$$

$$\underline{a}^1 = \underline{x}_2^1 - \underline{x}_1^1$$

$$\underline{n}^1 = \frac{1}{\|\underline{a}^1\|} \begin{bmatrix} a_y^1 \\ -a_x^1 \end{bmatrix}$$

$$\underline{x}^2 = N_1^2(\xi^2) \cdot \underline{x}_1^2 + N_2^2(\xi^2) \cdot \underline{x}_2^2$$

$$\underline{a}^2 = \underline{x}_2^2 - \underline{x}_1^2$$

$$\underline{n}^2 = \frac{1}{\|\underline{a}^2\|} \begin{bmatrix} a_y^2 \\ -a_x^2 \end{bmatrix}$$

$$\xi^2 = (\underline{x}^1 - \underline{x}_1^2) \cdot \underline{a}^2 / (\underline{a}^2 \cdot \underline{a}^2)$$

$$(\underline{x}^1 - \underline{x}^2)^T \underline{n}^2 = g_N$$

$$\underbrace{(\underline{n}^2)^T \underline{\underline{H}}_c}_{\underline{\underline{B}}_c} \underline{x}_c = g_N$$

$$\underline{x}^1 - \underline{x}^2 = \underline{\underline{H}}_c \underline{x}_c$$

$$= \begin{bmatrix} N_1^1 & 0 & N_2^1 & 0 \\ 0 & N_1^1 & 0 & N_2^1 \end{bmatrix} \begin{bmatrix} \underline{x}_1^1 \\ \underline{x}_2^1 \end{bmatrix} - \begin{bmatrix} N_1^2 & 0 & N_2^2 & 0 \\ 0 & N_1^2 & 0 & N_2^2 \end{bmatrix} \begin{bmatrix} \underline{x}_1^2 \\ \underline{x}_2^2 \end{bmatrix}$$

$$= \begin{bmatrix} \overbrace{N_1^1 \quad 0 \quad N_2^1 \quad 0}^{\xi^1} & \overbrace{-N_1^2 \quad 0 \quad -N_2^2 \quad 0}^{\xi^2} \\ 0 \quad N_1^1 \quad 0 \quad N_2^1 & 0 \quad -N_1^2 \quad 0 \quad -N_2^2 \end{bmatrix} \begin{bmatrix} \underline{x}_1^1 \\ \underline{x}_2^1 \\ \underline{x}_1^2 \\ \underline{x}_2^2 \end{bmatrix}$$

If $g_N < 0$

$$p_N = -\epsilon g_N$$

$$\underline{F}_c = \int_{\Gamma_c} p_N \underline{\underline{B}}_c^T \, d\Gamma = w \, p_N \underline{\underline{B}}_c^T \|\underline{a}^1\|$$

$$w_1 = 0{,}5 \qquad \xi_1 = 1/2 + 1/2 \frac{1}{\sqrt{3}}$$

$$w_2 = 0{,}5 \qquad \xi_2 = 1/2 - 1/2 \frac{1}{\sqrt{3}}$$

$$\underline{\underline{K}}_c = w \epsilon \underline{\underline{B}}_c^T \underline{\underline{B}}_c \|\underline{a}^1\|$$